

自动控制原理 第十章作业:

skills check: 全部.

Exercises: E10.2, E10.5, E10.9, E10.14, E10.15

Problems: P10.1, P10.13, P10.26, P10.38

Advanced Problems: AP10.4, AP10.7

CDP10.1 DP10.9

skills check:

正误判断: 1. X 2. X 3. V 4. V 5. V

多项选择题 6d 7b 8d 9a 10b 11c 12a 13a 14b 15b

术语匹配题 a l g d j h k c m e i f b

E10.2

a. 特征方程为 $H G_c(s) G(s) = H \frac{400}{s(s+40)} (k_p + \frac{1}{s})$

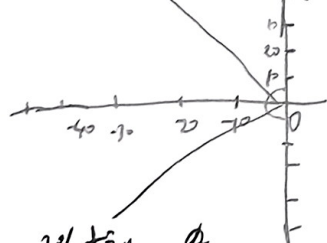
$$= H \frac{400(k_p s + 1)}{s^2(s+40)} = 0$$

$$\text{即 } H k_p \frac{400s}{s^3 + 40s^2 + 400} = 0$$

超调量为 20% 时 $\zeta = 0.45$.

由下根求 k_p 范围, $k_p = 0.5$.

闭环极点 $s_1 = -5$, $s_{2,3} = -2.7 \pm j2$



(b) 2% 标准

$$t_s = \frac{4}{\zeta \omega_n}$$

$$\text{近似} = \frac{4}{\zeta \omega_n} \quad s^2 + 40s + 400 k_p = 0$$

$$s^2 + 2(\zeta \omega_n)s + \omega_n^2 = 0$$

$$\zeta = 0.45 \quad \omega_n^2 = 400 k_p$$

$$\text{代入 } t_s = \frac{4}{0.45 \times 20} = 20.25$$

E10.5

$$\text{开环传递函数 } G_o(s) = G(s) G_c(s) = \frac{k}{s(s+1)(s+13)} \cdot \frac{s+13}{s+13}$$

特征方程

$$s(s+1)(s+13) + k(s+13) = 0$$

$$\text{展开 } s^3 + 14s^2 + 13s + (13s + k) = 0$$

$$\omega_n = 2, \quad \zeta = 0.5$$

$$s = -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2} = -1 \pm j1.6$$

近似

$$(s+1.1 + j1.67)(s+1.1 - j1.67)(s+a)(s+b) =$$

与原方程比较

$$s^4 + (a+2.2)s^3 + (2.2a+4+b)s^2 + (2.2b+4a)s + 4b =$$

$$\begin{cases} a+2.2=2 \\ 2.2a+4+b=11 \\ 2.2b+4a=19.5+k \\ 4b=27k \end{cases}$$

$$a=18.8, \quad b=73.64, \quad k = \frac{27 \times 73.64}{4} \approx 42$$

验证 $k = 42$ 代入其符合

E10.9

$$\text{开环传递函数 } G(s) = G_c(s) \cdot G(s) = \frac{k_1(s + \frac{k_2}{s})}{s(s+1)}$$

$$k_v = \lim_{s \rightarrow 0} s G(s) \cdot G(s) = k_2, \quad \text{选择 } k_2 = 5$$

$$\text{特征方程 } s^2 + (k_1 + 1)s + k_2 = 0$$

$$\text{对比 } s^2 + 2\zeta \omega_n s + \omega_n^2 = 0$$

$$\text{可得 } \omega_n = \sqrt{k_2} = \sqrt{5}$$

想使 P.O. = 5%, 要求 $\zeta = 0.69$

$$\text{因此 } 2\zeta \omega_n = k_1 + 1, \quad k_1 = 2.08$$

$$k_1 = 2.08, \quad k_2 = 5, \quad \text{P.O.} = 5\%$$

由于零点 $s = -1.08$ 处, 提高增益 $k_1 = 2$

则 P.O. = 5%

其阶跃响应见图。



E10.14

$$G_c(s) \cdot G(s) = \frac{5(7.5s+1)}{s(s+1)(0.25s+1)(110s+1)}$$

相位裕量: $\gamma = 180^\circ + \angle G_c(j\omega_c)$

$$\angle \frac{1}{j\omega} = -90^\circ \quad \angle \frac{1}{j\omega+1} = -\arctan(\omega)$$

$$\angle \frac{1}{0.25s+1} = -\arctan(0.25\omega)$$

$$\angle G_c(j\omega) = -90^\circ - \arctan(\omega) - \arctan(0.25\omega)$$

γ : 在 $\omega=1 \text{ rad/s}$ 处增益为 0dB

ω : 转折频率 1 rad/s $0^\circ \rightarrow -90^\circ$

$\frac{1}{0.25s+1}$ 转折频率 4 rad/s

增益裕量 $G_{(s)} = \frac{147.5}{147.5}$

$\gamma = 45^\circ$ 假设 $\omega_c > \omega_b$

$$180^\circ - 45^\circ = 135^\circ$$

设 $\omega_c' = 1.1 \text{ rad/s}$

$$\angle G_c(j\omega) = -90^\circ - 135^\circ = -180^\circ$$

$$\omega_b = 7.5 \quad T_1 = \frac{7.5}{7.5} \approx 6.2857$$

$$T_2 = 14.678 \quad 6.82 \approx 92.57$$

增益裕量 $G_{(s)} = \frac{147.5}{147.5}$

E10.15 期望的穿越频率为 $\omega_c = 10 \text{ rad/s}$

$$20 \log |G_c(j\omega) G(j\omega)| = -81 \text{ dB}$$

$$\angle (G_c(j\omega) G(j\omega)) = -169^\circ$$

因此相角裕度 $P.M. = 11^\circ$

$$\phi = 3^\circ - 11^\circ = -8^\circ \quad m = 8.1 \text{ dB}$$

由于 $\phi > 0$, $m > 0$, 因此需要超前网络

校正原有系统

P10.1

(a) 开环传递函数 $L(s) = G_c(s) G(s) H(s) = \frac{(H_1 T_1) k_1 k_2}{s^2 (H_2 T_2) (J s^2)}$

期望 $\zeta = 0.6$ $T_s \leq 2.5$, 即 $\omega_n > 1.6$

校正: $T(s) = \frac{k}{s^2 + k}$ $k = k_1 k_2 / J$, $k = \omega_n^2$

选择 $k = 20$ $\omega_n > 1.6$

$$G_c(s) H(s) = \frac{20}{s^2} \text{ 的伯德图}$$

校正系统相角裕量为 0°

在 ω_c 处加相角为 40°

$$\sin 40^\circ = \frac{\alpha - 1}{\alpha + 1} = 0.64$$

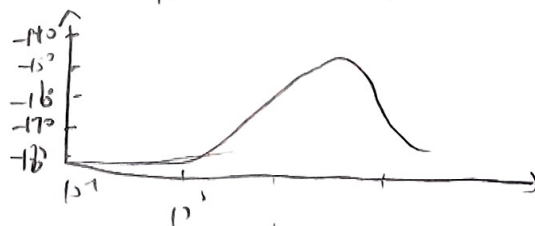
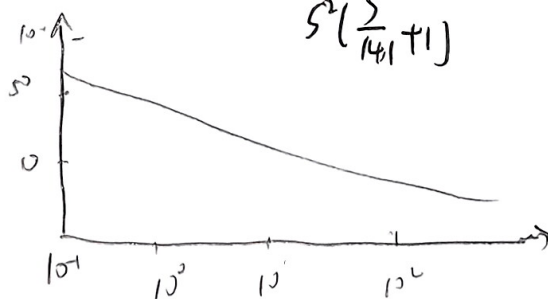
$$\alpha = 4.6$$

$$20 \log \alpha = 20 \log 4.6 = 6.63 \text{ dB}$$

$$\omega_n = 6.6 \text{ rad/s} \quad P.M. = \omega_n \sqrt{2} = 4.1$$

$$\xi = \frac{P}{\omega_n} = 3.07$$

$$G_c(s) G(s) H(s) = \frac{20(\frac{s}{3.07} + 1)}{s^2(\frac{s}{14.1} + 1)}$$



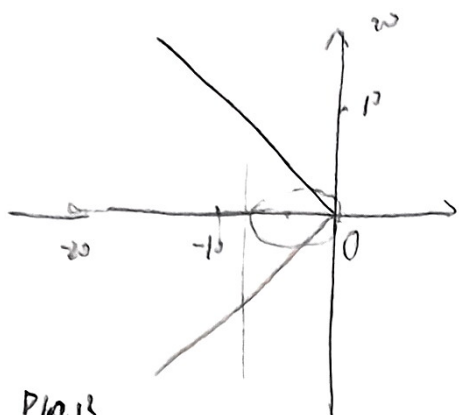
(b) 由于 $\omega_n > 1.6$ 补偿器零点 $\xi = 0.6$

选择 $P = 20$ 设置 $G_c(s) = \frac{s+1.6}{s+20}$

为满足 $\zeta = 0.6$, $k = 200$

$$G_c(s) G(s) H(s) = \frac{250(s+1.6)}{s^2(s+20)} = \frac{20(\frac{s}{1.6} + 1)}{s^2(\frac{s}{20} + 1)}$$





P10.13

$$G(s) = G_c(s) \cdot \frac{8}{s(s+2)(s+4)} = \frac{8K}{s(s+2)(s+4)}$$

$$\angle G(j\omega) = -180^\circ + 4^\circ = -135^\circ$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{4}\right)$$

$$\text{At } \angle G(j\omega) = -135^\circ$$

$$-90^\circ - \tan^{-1}\left(\frac{\omega_c}{2}\right) - \tan^{-1}\left(\frac{\omega_c}{4}\right) = -135^\circ$$

$$\tan^{-1}\left(\frac{\omega_c}{2}\right) + \tan^{-1}\left(\frac{\omega_c}{4}\right) = 45^\circ$$

$$\omega_c \approx 1.23 \text{ rad/s}$$

$$\omega_c = 1.23 \text{ rad/s} \quad |G(j\omega_c)| = 1$$

$$K \approx 1.338$$

$$\omega_B = 1.5\omega_c = 1.84 \text{ rad/s}$$

$$(b) \quad K_v = \lim_{s \rightarrow 0} s G(s) = \lim_{s \rightarrow 0} \frac{8K}{s+2} = 4K$$

$$\omega_{c0} = 0.894 \text{ rad/s}$$

$$-90^\circ - \tan^{-1}(0.447) - \tan^{-1}(0.224) = -126.7^\circ$$

$$P.M. \text{ margin} = 180^\circ - 126.7^\circ = 53.3^\circ$$

$$G_c(s) = \frac{Ts+1}{Ts+1}$$

$$\omega_c = 0.848 \text{ rad/s}$$

$$20 \log |G(j\omega)| = -20 \log |1 + j\omega| = -0.52 \text{ dB}$$

$$\delta G_c(s) = \frac{11.8s+1}{12.5s+1}$$

$$\beta T = 0.0799 \text{ rad}$$

$$\text{验证 } K_v: \text{通过 } G_c(s) = \frac{11.8s+1}{12.5s+1} \text{ 校正后相角 } \angle G_c(j0.848) = -0.3^\circ$$

$$\angle G(j0.848) = -125^\circ$$

$$P.M. = 180^\circ - 125.1^\circ = 54.7^\circ$$

P10.26

$$G(s) = \frac{0.2}{s(s+1)(s+2)}$$

$$|G(j\omega)| = \frac{0.2}{\omega \sqrt{1+\omega^2} \sqrt{4+\omega^2}}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\omega - \tan^{-1}(2\omega)$$

$$|G(j\omega_g)| = 1 \quad P.M. = 180^\circ + \angle G(j\omega_g)$$

$$G_c(s) = K \frac{s+2}{s+1}$$

$$\omega_B = 1.5\omega_c \quad \omega_c \approx \frac{0.75}{1.5} = 0.5 \text{ rad/s}$$

$$\omega_c = 0.5 \text{ rad/s} \quad |G(j0.5)| = \left| \frac{0.2}{j0.5(1+j)(1+j2)} \right|$$

$$\omega_c = 0.5 \text{ rad/s} \text{ 成为增益穿越频率}$$

$$P.M. \text{ margin} = 180^\circ - \angle G(j0.5) = 180^\circ - 133^\circ = 47^\circ$$

$$P.M. \geq 45^\circ \quad \phi_m = 13 - (-8.2^\circ) + 5^\circ = 43.2^\circ$$

$$\sin \phi_m = \frac{d-1}{d+1} \quad d=5 \quad n = \frac{T}{T_d} = \omega_c = 0.5 \quad T = 0.894$$

$$\text{校正器 } K = \frac{1+4.47}{1+0.894} = 3.96 \quad \frac{(1+4.47)}{1+0.894}$$

$$\text{验证: } \omega_c = 0.5 \text{ rad/s} \text{ 时 } |G(j\omega_c)| = 1, P.M. = 53.3^\circ > 45^\circ$$

P10.38.

$$\omega_B = \omega_c 1.5 = 0.75 \text{ rad/s}$$

超前校正网络:

$$G_c(s) = \frac{s+2}{s+1}$$

$$\phi_m = \arccos(\omega_c) - \arccos(\omega_p)$$

在 $\omega_c = 1.0 \text{ rad/s}$ 时, 系统相角裕度达到 30°

$$|G(j\omega)| = \left| \frac{40}{j\omega(1+j\omega)} \right|$$

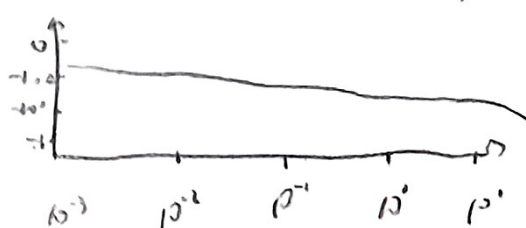
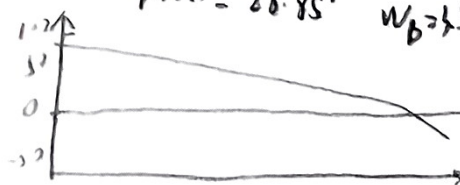
$$\angle G(j\omega) = -90^\circ - \arccos\left(\frac{\omega}{2}\right)$$

$$G_c(s) = \frac{1+1.48s}{1+1.08s}$$

$$\text{由于 } K_v = 20,$$

$$\omega_c = 2 \text{ rad/s}$$

$$P.M. = 28.85^\circ \quad \omega_B = 3.57 \text{ rad/s}$$



AP10.4

前向通路 \$k_1 \frac{\delta}{s(0.5s+1)}\$ 速度反馈 \$k_2 s\$

$$\text{开环 } k_1 \frac{\delta}{s(0.5s+1)} (1+k_2 s) = \frac{8k_1 (1+k_2 s)}{s(0.5s+1)}$$

$$\text{闭环 } \frac{Y(s)}{R(s)} = \frac{8k_1 (1+k_2 s)}{0.5s^2 + s + 8k_1 k_2 s + 8k_1}$$

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad \omega_n^2 = 16k_1 \quad \omega_n = 4\sqrt{k_1}$$

$$2\zeta\omega_n = 2 + 16k_1 k_2 \quad \zeta = \frac{1+8k_1 k_2}{4\sqrt{k_1}}$$

$$T_s < 0.55 \text{ (2\% 误差)}$$

$$T_s = \frac{4}{\zeta\omega_n} < 0.55 \Rightarrow \zeta\omega_n > 7.27$$

超调量 D.O. $\leq 5\%$

$$D.O. = 100 e^{-\frac{\zeta}{\sqrt{1-\zeta^2}}} \leq 5\% \Rightarrow \zeta > 0.59 \text{ 取 } \zeta = 0.6$$

$$\zeta\omega_n > 7.27 \quad \omega_n \frac{\zeta}{0.6} = 12.35 \text{ rad/s} \quad 16k_1 = \omega_n^2 = 196 \quad \sqrt{k_1} = 12.25$$

$$\text{取 } k_2 = \frac{7.4}{98} = 0.075 \text{ 验证 } \omega_n = 4\sqrt{16} = 16 \text{ rad/s}$$

$$\text{AP10.7} \quad 9.0 = 100 e^{-\frac{\zeta\omega_n}{\sqrt{1-\zeta^2}}} = 9.5\% \Rightarrow \zeta = \frac{1+8 \times 0.075 \times 12.25}{4} = 0.476$$

\$G(s) = 1, k=311\$ 闭环传递函数

$$T(s) = \frac{k(s+5)}{s^4 + 38s^3 + 296s^2 + (448s + 311)}$$

\$k=311\$ 时特征方程

$$s^4 + 38s^3 + 296s^2 + 259s + 91 = 0$$

\$\zeta = \frac{1}{\sqrt{2}}\$ 临界阻尼

$$s = -1.618 \pm j1.617, \quad s = -6.25, \quad s = -28.51$$

(a) \$G(s) = 1\$

$$P.O. = 6.5 \quad T_s = 2.5s \quad T_r = 1.6s$$

$$\text{滤波器 } G_F(s) = \frac{3}{s+1}, \quad k=311$$

$$\text{闭环 } D.O. = 3.9\% \quad T_s = 2.8s \quad T_r = 1.3s$$

$$\text{(b) 滤波器 } G_F(s) = \frac{1.8}{s+1.8}$$

$$\text{其阶跃响应 } D.O. = 0.7\%$$

$$T_s = 2.14s, \quad T_r = 1.3s$$

CDP 10.1

$$\frac{\theta(s)}{V(s)} = \frac{26.035}{s(s+33.142)}$$

$$s^2 + (33.142 + 26.035k_0)s + 26.035k_p = 0$$

取 + 阶跃响应 = 阶跃响应 24.82

$$\omega_n T_s = 4.82, \quad T_s < 0.75s$$

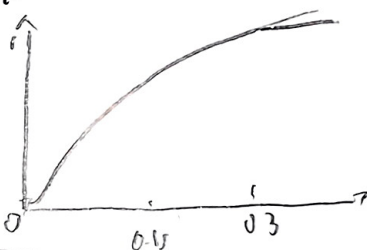
$$\text{选择 } \omega_n = 19.28$$

使实际特征方程与期望相符

$$s^2 + \omega_n \zeta s + \omega_n^2 = s^2 + (33.142 + 26.035k_0)s + 26.035k_p$$

$$\text{求 PD 控制器 } G_C(s) = 14.28 + 0.0125s$$

$$T_s = 0.24s$$



DP 10.9

$$\text{PID 控制器 } G_C(s) = k_p + k_d s + \frac{k_i}{s} = \frac{1.554s^2 + 1.08s + 1}{s}$$

$$G_C(s) = K \left(\frac{s\omega_c}{s+\omega_b} \right) \left(\frac{s\omega_c}{s+\omega_d} \right) = 6.02 \left(\frac{s(1.5)}{s+0.667} \right) \left(\frac{s(1.5)}{s+0.667} \right)$$

两者都在 \$T=0.1s\$ 时系统的响应下稳定

PID 控制器 超前裕度 P.M. = 40°

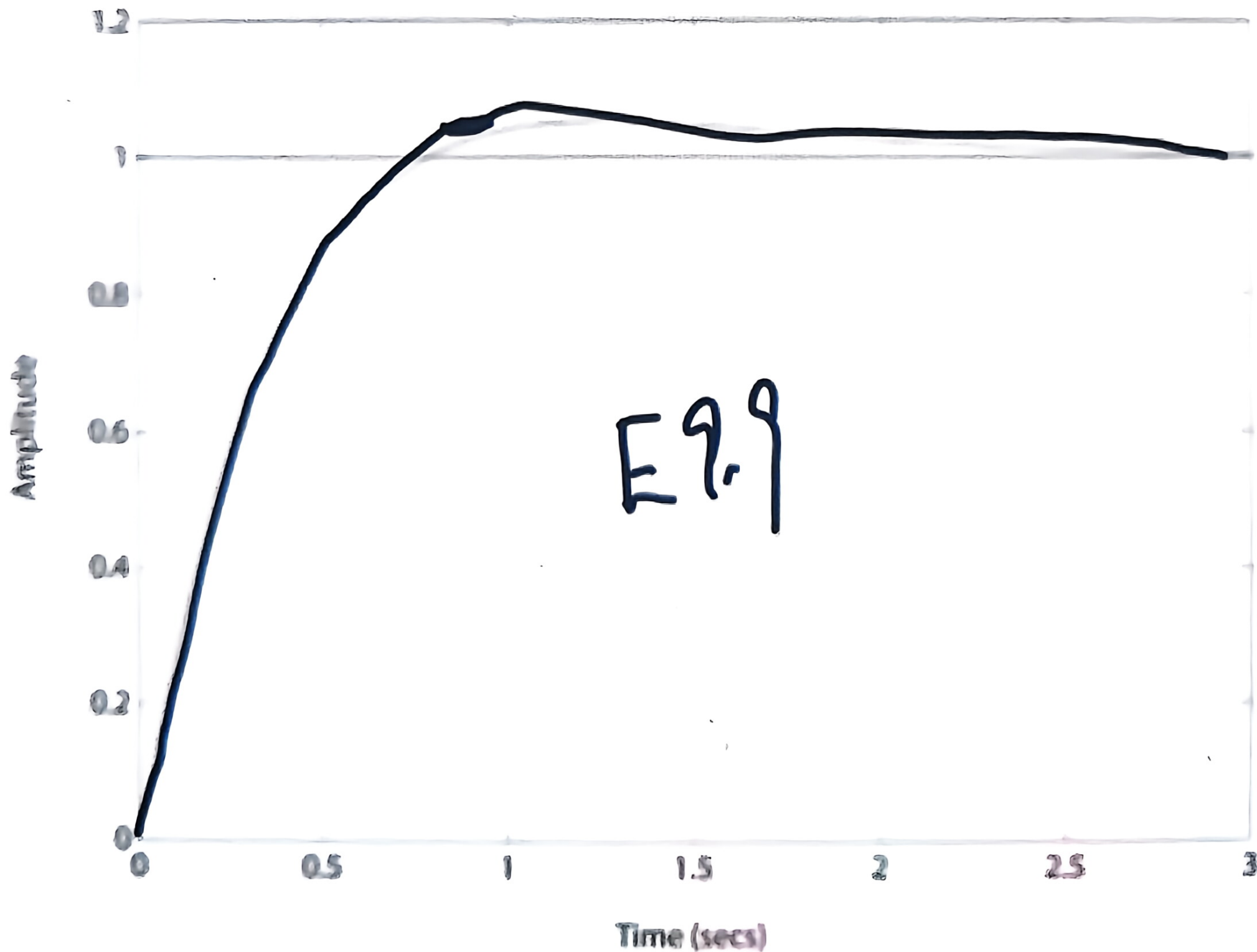
对超前滞后校正网络 P.M. = 45°

超前校正更能保持稳定

对 \$T=0.25s\$ 时

超前滞后校正 P.M. = 20° 而 PID 控制器不稳定





CDP10-1

